

# Package ‘MultFourier’

September 21, 2026

**Type** Package

**Title** Computes the p-values for Multinomial Goodness-of-fit tests

**Version** 1.0

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**Description** Computes the p-value for an exact Multinomial, Pearson's Chi-Squared, log-likelihood ratio, or Power-Divergence tests. The main function, `pval_fourier`, uses a Fourier series expansion. Also, the library provides an exhaustive method `pval_exhaustive`, which evaluates all elements within the support of the Multinomial distribution. The function `pval_flexible` uses either of these methods, depending on the size of the instance.

**License** GPL (>= 2)

**Encoding** UTF-8

**Suggests** knitr,  
rmarkdown,  
ggplot2

**VignetteBuilder** knitr

**Config/roxygen2/version** 8.1.0

## Contents

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## Description

Computes the p-value for an exact Multinomial, Pearson's Chi-Squared, log-likelihood ratio, or Power-Divergence tests. The main function, `pval_fourier`, uses a Fourier series expansion. Also, the library provides an exhaustive method `pval_exhaustive`, which evaluates all elements within the support of the Multinomial distribution. The function `pval_flexible` uses either of these methods, depending on the size of the instance.

## Details

The `MultFourier` package provides efficient methods for computing p-values in multinomial tests, including both exact and approximate approaches. The exact p-value is computed using an exhaustive algorithm, which is recommended for small datasets. For larger datasets, an approximation is provided using a Fourier series algorithm, which is computationally efficient and provides control over precision through parameters such as `max_terms`, `rel_eps`, and `undersampling`.

The package includes the following methods:

- `pval_exhaustive`: Computes the exact p-value by enumerating all possible outcomes. Suitable for small datasets.
- `pval_fourier`: Approximates the p-value using a Fourier series expansion. Efficient for large datasets.
- `pval_flexible`: Automatically selects between the exhaustive and Fourier series methods based on the dataset size.

Key features:

- Control over precision through parameters such as `max_terms`, `rel_eps`, and `undersampling`.
- Support for multiple test statistics, including logP, Pearson's chi-square, and log-likelihood ratio.
- Verbose output for debugging and analysis.

## Author(s)

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Authors:

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- Charles Thraves
- Felipe Subiabre

## References

- Thraves, C. and Subiabre, F. et al. (Year): "Title of the Paper". Journal Name, Volume(Issue), Pages.

**See Also**

- Report bugs at <https://github.com/PabRvss/MultFourier/issues>

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|                   |                                             |
|-------------------|---------------------------------------------|
| print.multfourier | <i>Print method for multfourier objects</i> |
|-------------------|---------------------------------------------|

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**Description**

Print method for multfourier objects

**Usage**

```
## S3 method for class 'multfourier'
print(x, ...)
```

**Arguments**

|     |                                                                                            |
|-----|--------------------------------------------------------------------------------------------|
| x   | Object of class "multfourier" returned by pval_fourier, pval_exhaustive, or pval_flexible. |
| ... | Additional arguments (currently unused).                                                   |

**Value**

Invisibly returns the input object.

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|                 |                                                                      |
|-----------------|----------------------------------------------------------------------|
| pval_exhaustive | <i>Exact multinomial p-value computation via iterative bisection</i> |
|-----------------|----------------------------------------------------------------------|

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**Description**

Exact multinomial p-value computation via iterative bisection

**Usage**

```
pval_exhaustive(
  x,
  p = rep(1/length(x), length(x)),
  stat = c("llr", "pd", "chi2", "pmf"),
  lambda = 1,
  n_threads = 1,
  max_time = NULL,
  verbose = FALSE,
  ...
)
```

**Arguments**

|                  |                                                                                              |
|------------------|----------------------------------------------------------------------------------------------|
| <b>x</b>         | Numeric vector of observed counts for each category.                                         |
| <b>p</b>         | Numeric vector of theoretical probabilities under the null hypothesis.                       |
| <b>stat</b>      | Test statistic: "llr", "pd", "chi2", or "pmf". Default is "llr".                             |
| <b>lambda</b>    | Real parameter for Power Divergence when stat = "pd". Default is 1. Ignored if stat != "pd". |
| <b>n_threads</b> | Number of execution threads (integer >= 1). Default is 1.                                    |
| <b>max_time</b>  | Maximum execution time in seconds (optional).                                                |
| <b>verbose</b>   | Logical; if TRUE, prints progress information. Default is FALSE.                             |
| <b>...</b>       | Additional internal parameters passed to the C engine.                                       |

**Value**

Returns an S3 object of class "multfourier" with the following attributes:

- x** The input vector of observed realizations for each category.
- p** The input vector of probabilities for each category under the null hypothesis.
- pval** The computed p-value.
- stat** A string describing the test statistic used.
- time** The total execution time of the algorithm in seconds.
- status** The final status ID of the algorithm upon completion (0: Converged, 1: Maximum time reached).
- message** The finishing status displayed as a human-readable message.
- method** A string with value "exhaustive".

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|               |                                                            |
|---------------|------------------------------------------------------------|
| pval_flexible | <i>Multinomial p-value computation via flexible method</i> |
|---------------|------------------------------------------------------------|

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**Description**

Evaluates whether the dimension of the multinomial support allows exact enumeration (iterative bisection) or routes to the Fourier inversion series method.

**Usage**

```
pval_flexible(
  x,
  p = rep(1/length(x), length(x)),
  stat = c("llr", "pd", "chi2", "pmf"),
  lambda = 1,
  undersampling = 1,
  rel_eps = 0.001,
  B = 30,
```

```

    max_terms = 10000,
    max_time = NULL,
    avg_window = 0,
    avg_flat = FALSE,
    n_threads = min(2L, parallel::detectCores()),
    precision = c("double", "double-double"),
    precompute = TRUE,
    verbose = FALSE,
    ...
)

```

### Arguments

|               |                                                                                                                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x             | Numeric vector of observed counts for each category.                                                                                                                                                                            |
| p             | Numeric vector of theoretical probabilities under the null hypothesis (must sum to 1).                                                                                                                                          |
| stat          | Test statistic: "llr" (Log-Likelihood Ratio), "pd" (Power Divergence), "chi2" (Pearson's Chi-squared), or "pmf" (Multinomial Point Probability Mass). Default is "llr".                                                         |
| lambda        | Real parameter for Power Divergence when stat = "pd". Default is 1. Ignored if stat != "pd".                                                                                                                                    |
| undersampling | Undersampling factor (> 0) for the period in the Fourier series. Default is 1.                                                                                                                                                  |
| rel_eps       | Relative convergence tolerance (>= 0). Default is 0.001.                                                                                                                                                                        |
| B             | Integer >= 0; number of consecutive terms within tolerance required for convergence. Default is 30.                                                                                                                             |
| max_terms     | Maximum number of harmonic terms to evaluate in the series (positive integer). Default is 10000.                                                                                                                                |
| max_time      | Maximum execution time in seconds (optional).                                                                                                                                                                                   |
| avg_window    | Fraction of final partial sums to average (between 0 and 1). Default is 0.                                                                                                                                                      |
| avg_flat      | Logical; if TRUE, uses a flat moving average; if FALSE, applies linear weighting towards the end. Default is FALSE.                                                                                                             |
| n_threads     | Number of execution threads (positive integer >= 1). Default is min(2L, parallel::detectCores()).                                                                                                                               |
| precision     | Arithmetic precision: "double" (standard) or "double-double" (extended precision).                                                                                                                                              |
| precompute    | Logical; if TRUE, uses precomputed transforms for acceleration. Default is TRUE.                                                                                                                                                |
| verbose       | Logical; if TRUE, prints progress information. Default is FALSE.                                                                                                                                                                |
| ...           | Additional internal parameters passed to the C engine (e.g. enum_cutoff = 7.4, the threshold for log <sub>10</sub> support size under which exact bisection is selected, corresponding to approximately 1 second of execution). |

**Value**

Returns an S3 object of class "multfourier" with the following attributes:

- x** The input vector of observed realizations for each category.
- p** The input vector of probabilities for each category under the null hypothesis.
- pval** The computed p-value.
- Time\_gamma** The time spent optimizing gamma in seconds.
- W** The effective width of the distribution support interval.
- stat** A string describing the test statistic used (e.g. "Log-Likelihood Ratio", "Pearson's Chi-Square", "Probability Mass Function", or "power-divergence (lambda = ...)").
- Gamma** The optimal gamma obtained in the first part of the method.
- n\_terms** The number of terms of the Fourier sum evaluated.
- time** The total execution time of the algorithm in seconds.
- status** The final status ID of the algorithm upon completion:
  - 0: Converged.
  - 1: Maximum time reached.
  - 2: Maximum number of terms reached.
  - 3: Could not solve the optimization of gamma.
  - 4: Numerical instability detected (magnitude explosion).
- message** The finishing status displayed as a human-readable message.
- method** A string with value "fourier" or "exhaustive".

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pval\_fourier

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*Multinomial p-value computation via Fourier series inversion*


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**Description**

Multinomial p-value computation via Fourier series inversion

**Usage**

```
pval_fourier(
  x,
  p = rep(1/length(x), length(x)),
  stat = c("llr", "pd", "chi2", "pmf"),
  lambda = 1,
  undersampling = 1,
  rel_eps = 0.001,
  B = 30,
  max_terms = 10000,
  max_time = NULL,
  avg_window = 0,
```

```

    avg_flat = FALSE,
    n_threads = min(2L, parallel::detectCores()),
    precision = c("double", "double-double"),
    precompute = TRUE,
    verbose = FALSE,
    ...
)

```

### Arguments

|                      |                                                                                                                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>x</b>             | Numeric vector of observed counts for each category.                                                                                                                                                                            |
| <b>p</b>             | Numeric vector of theoretical probabilities under the null hypothesis (must sum to 1).                                                                                                                                          |
| <b>stat</b>          | Test statistic: "llr" (Log-Likelihood Ratio), "pd" (Power Divergence), "chi2" (Pearson's Chi-squared), or "pmf" (Multinomial Point Probability Mass). Default is "llr".                                                         |
| <b>lambda</b>        | Real parameter for Power Divergence when stat = "pd". Default is 1. Ignored if stat != "pd".                                                                                                                                    |
| <b>undersampling</b> | Undersampling factor (> 0) for the period in the Fourier series. Default is 1.                                                                                                                                                  |
| <b>rel_eps</b>       | Relative convergence tolerance (>= 0). Default is 0.001.                                                                                                                                                                        |
| <b>B</b>             | Integer >= 0; number of consecutive terms within tolerance required for convergence. Default is 30.                                                                                                                             |
| <b>max_terms</b>     | Maximum number of harmonic terms to evaluate in the series (positive integer). Default is 10000.                                                                                                                                |
| <b>max_time</b>      | Maximum execution time in seconds (optional).                                                                                                                                                                                   |
| <b>avg_window</b>    | Fraction of final partial sums to average (between 0 and 1). Default is 0.                                                                                                                                                      |
| <b>avg_flat</b>      | Logical; if TRUE, uses a flat moving average; if FALSE, applies linear weighting towards the end. Default is FALSE.                                                                                                             |
| <b>n_threads</b>     | Number of execution threads (positive integer >= 1). Default is min(2L, parallel::detectCores()).                                                                                                                               |
| <b>precision</b>     | Arithmetic precision: "double" (standard) or "double-double" (extended precision).                                                                                                                                              |
| <b>precompute</b>    | Logical; if TRUE, uses precomputed transforms for acceleration. Default is TRUE.                                                                                                                                                |
| <b>verbose</b>       | Logical; if TRUE, prints progress information. Default is FALSE.                                                                                                                                                                |
| <b>...</b>           | Additional internal parameters passed to the C engine (e.g. enum_cutoff = 7.4, the threshold for log <sub>10</sub> support size under which exact bisection is selected, corresponding to approximately 1 second of execution). |

### Value

Returns an S3 object of class "multfourier" with the following attributes:

- x** The input vector of observed realizations for each category.
- p** The input vector of probabilities for each category under the null hypothesis.

**pval** The computed p-value.

**Time\_gamma** The time spent optimizing gamma in seconds.

**W** The effective width of the distribution support interval.

**stat** A string describing the test statistic used (e.g. "Log-Likelihood Ratio", "Pearson's Chi-Square", "Probability Mass Function", or "power-divergence (lambda = ...)").

**Gamma** The optimal gamma obtained in the first part of the method.

**n\_terms** The number of terms of the Fourier sum evaluated.

**time** The total execution time of the algorithm in seconds.

**status** The final status ID of the algorithm upon completion:

- 0: Converged.
- 1: Maximum time reached.
- 2: Maximum number of terms reached.
- 3: Could not solve the optimization of gamma.
- 4: Numerical instability detected (magnitude explosion).

**message** The finishing status displayed as a human-readable message.

**method** A string with value "fourier".

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